

# JIASEN LIU

(609)-874-4930  $\diamond$  jliu3566@usc.edu

## EDUCATION

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**University of Southern California (USC)**

*Bachelor of Science*; Mathematics; Computer Science

*Master of Arts*; Mathematics

GPA: 3.95/4.00

Aug 2022 - May 2026

Jun 2024 - May 2026

**Academic Interest:** Homotopy Theory, Algebraic Geometry, Algebraic Topology

**Relevant (graduate-level) Coursework:** Algebraic Topology, Algebra I & II, Commutative Algebra, Homological Algebra, Toric Algebraic & Symplectic Geometry, Smooth Manifolds, Enumerative Combinatorics, Real Analysis I, Advanced Analysis of Algorithms, Voting Theory.

## ACADEMIC EXPERIENCES

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**Research Assistant**, Theory Group, USC

Apr 2024 - Present

- Worked on a theoretical computer science project about justified committee selection under the supervision of professor David Kempe.

**Undergraduate Researcher**, PCMI & Institute for Advanced Study

Jul 2024 - Jul 2024

- Collaborated with 4 other undergraduates on a project about classifying spaces and principal bundles; presented the results to about 100 audiences.

**Mentee**, Directed Reading Course, USC

Jan 2024 - Jul 2024

- Studied  $\infty$ -category theory and related topics under the supervision of professor Joseph Helfer; came up with a 30-page survey about topics covered.

**Research Assistant**, AutoDrive Lab, USC

Aug 2023 - Dec 2023

- Helped building a multi-agent driving simulator to collect data for imitation learning algorithms under the supervision of professor Rahul Jain.

**Undergraduate Researcher**, Polymath Jr. REU, Williams College

Jun 2023 - Aug 2023

- Collaborated with 3 other undergraduates on an analytical number theory project about Diophantine equations under the supervision of professor Steven Miller and professor Hùng Viet Chu.

## WORKSHOPS & SEMINARS

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**Beijing Inst. of Math. Sciences & Applications: Algebraic Curves**

Sep 23 - Dec 18, 2024

**Preliminary Arizona Winter School: Local Fields**

Sep 23 - Nov 22, 2024

**USC CS Theory Group (Lunch) Seminar**

Aug 2024 - Present

**IAS/PCMI Undergrad. Summer School: Motivic Homotopy Theory**

Jul 7 - Jul 27, 2024

**Connecticut Summer School in Number Theory**

Jun 9 - Jun 16, 2024

**USC Combinatorics Seminar: Asymptotic Algebraic Combinatorics**

Jan 2024 - Apr 2024

## TEACHING EXPERIENCES

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**Teaching Assistant**, USC

Aug 2023 - Present

- Fall 2024: MATH 430 (Number Theory), CSCI 270 (Algorithms & Theory of Computing)
- Spring 2024: CSCI 170 (Discrete Mathematics)
- Fall 2023: MATH 425a (Mathematical Analysis I)

**Counselor**, Program in Mathematics for Young Scientists, Boston University    Jun 2023 - Aug 2023

- Critiqued and graded the daily problem sets of 5 high school students, gave presentations on relevant topics to around 100 audiences, and supervised a 5-student research group.

## **AWARDS & SCHOLARSHIPS**

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**Archer Undergraduate Research Award**, USC

Feb 2024